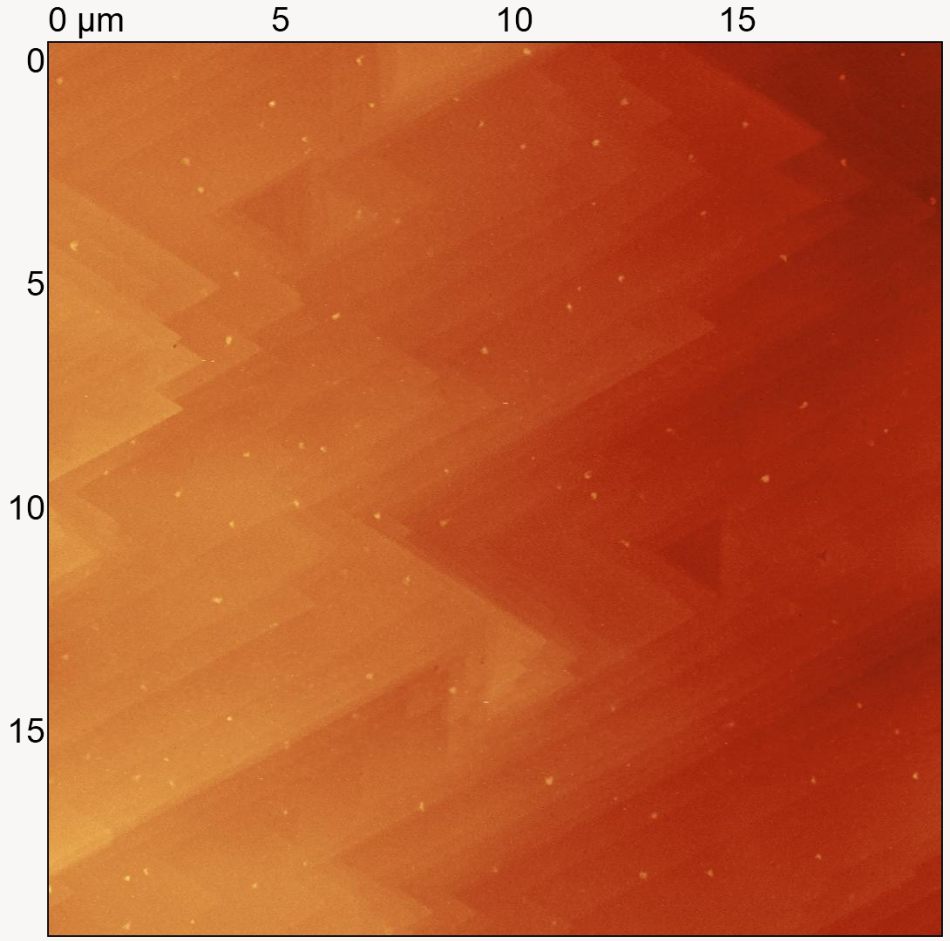
Epitaxial Sb and Al thin films on III-V(111)A substrates via low-overhead growth processes



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Jimmy Rushing, Asmaa Zoheiri, Merve Baksi, Nahian Nawar, John Mcelearney, Pan (Thachachanok) Menasuta, Javad Shabani(NYU), Darryl Shima (UNM), Ganesh Balakrishnan (UNM), Fabian Hartman (JMU), Selena Maldonado(JMU), Manuel Meyer (JMU), Sven Höfling (JMU), Tom Vandervelde, Paul Simmonds

Department of Electrical and Computer Engineering
Tufts University

Why (111)? All three materials share a triangular surface symmetry

III-Vs

GaSb, AlSb, InAs, GaAs

BULK STRUCTURE
zinc blende

SPACE GROUP
F-43m (#216)

IDEAL (111) PLANE
polar A/B triangular net; 3-fold surface symmetry

IDEAL (001) PLANE
nominal square net; III/V termination gives C2v-like surface symmetry

Al

bulk aluminum metal

BULK STRUCTURE
face-centered cubic

SPACE GROUP
Fm-3m (#225)

IDEAL (111) PLANE
close-packed triangular metal plane; 3-fold surface symmetry

IDEAL (001) PLANE
square fcc surface net; 4-fold C4v-like surface symmetry

Sb

bulk rhombohedral A7 semimetal

BULK STRUCTURE
A7 rhombohedral

SPACE GROUP
R-3m (#166)

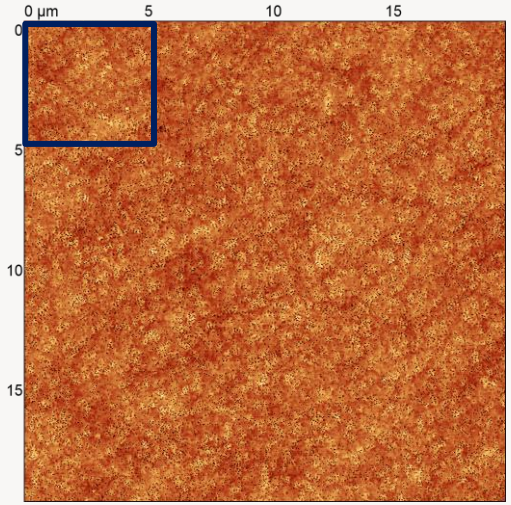
IDEAL (111) PLANE
natural buckled-bilayer surface; 3-fold surface symmetry

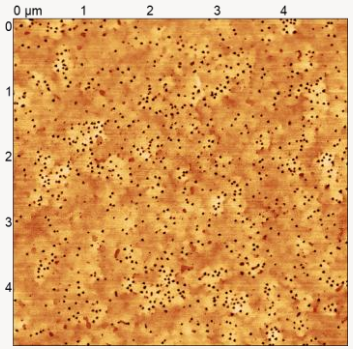
IDEAL (001) PLANE
not a cubic analog; low-symmetry rhombohedral cut, not the natural bilayer plane

AFM: full stacks with sub-nanometer-smooth surfaces

Sample 7

AlOx ~2.9 nm
Al 25.2 nm
Sb 1.38 nm
AlSb ~55 nm
GaSb(111)A

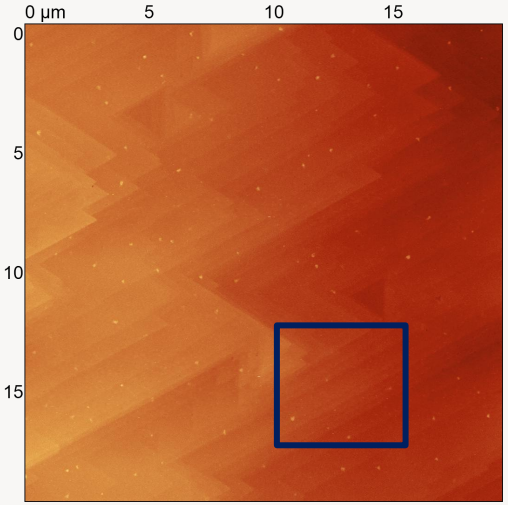


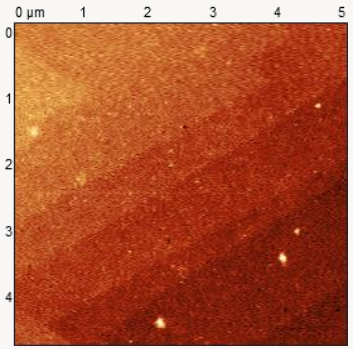


5 x 5 um inset - RMS 0.863 nm

Sample 8

AlOx X nm
Al 3.5 nm
Sb 10 nm
AlSb 40 nm
GaSb(111)A

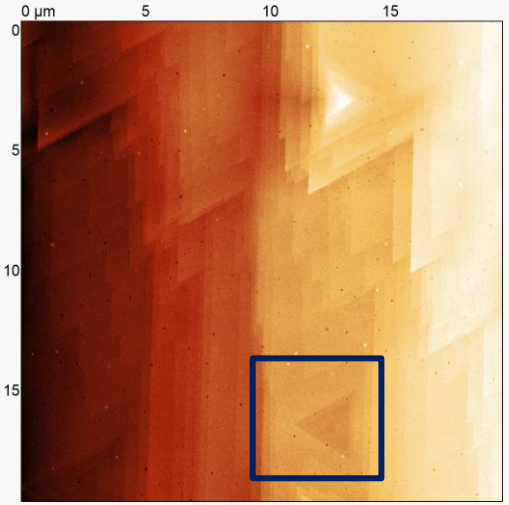


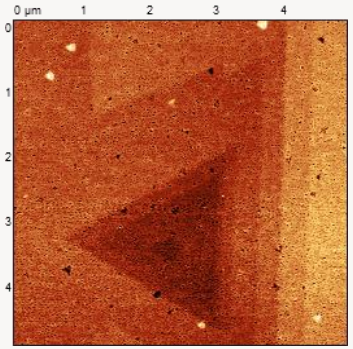


5 x 5 um inset - RMS 0.747 nm

Sample 9

AlOx X nm
Al ~3.5 nm
Sb 20 nm
AlSb 40 nm
GaSb(111)A





5 x 5 um inset - RMS 0.781 nm

Surface steps of ~0.20 nm remain measurable through the native oxide; Al(111) monolayer: 0.234 nm.

The sidecar points to an Sb/Al III-V(111)A platform

What we have shown

Crystalline Sb(111), 1.5-42 nm, on GaSb(111)A-based AlSb/GaSb templates. Al(111), 5-25 nm, on 1.5-30 nm Sb.

Possible implication

If Al(111) conforms to thin Sb across this range, Sb may be a general adapter for epitaxial Al(111) on III-V(111)A.

Future work

Regrow III-Vs on the epitaxial Al contact layer.

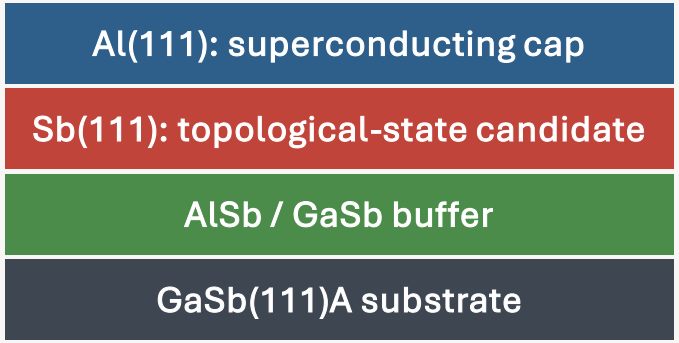
Test whether Al grows this well directly on III-Vs, without Sb.

Bring III-Vs back onto Sb; GaSb-on-Sb growths are underway.

Measure low-temperature transport in epitaxial Al.

XPS to see what the oxide is.

Test A7 alternatives at both interfaces. Bi normally needs more process overhead.



Low-overhead Sb/Al growth on AlSb/GaSb(111)A: ultrathin Sb for topology studies or, potentially, an ultrathin topologically inert layer that conforms Al to III-V stacks.